



Microphone

1925**Microphone**

The horn is replaced by electric microphones, in which vibrations move electromagnets to create a varying electrical signal that moves the needle to make the grooves.

1931**Reel-to-reel tape recorder**

German Fritz Pfleumer invents the reel-to-reel tape recorder, in which the fluctuations in the electrical signal are recorded in the magnetic coating of moving tape. The AEG company develops it into the Magnetophone.

Magnetophone

**1978****Personal stereo**

German-Brazilian Andreas Pavel's Stereobelt of 1972 plays cassette tapes through ear phones on a tiny, portable, battery-driven player. In 1978, Sony launches a portable music player—the Sony Walkman.



Sony Walkman

1898**Telegraphone**

Danish engineer Valdemar Poulsen's telegraphone is the first device to record and play back sound magnetically, using a wire wrapped around a cylinder to record the varying magnetic fields created by sound vibrations.



Telegraphone

1948**Record player**

Long-playing vinyl records, playing at 33 and 45 revolutions per minute (rpm) instead of 78 rpm, are introduced. They give much longer playing times and the sound quality is better.

1950s record player

**1982****Compact disks**

CDs digitally store large amounts of sound data that can be read by a laser. They quickly replace large, easily scratched vinyl records.

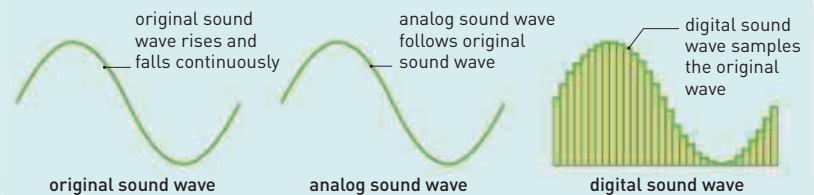
1999**MP3 players**

These use digital recordings stored as computer data, enabling music to be downloaded, or swapped from computer to personal player instantly.



ipod nano

apparatus support

**SOUND WAVES**

Sound is created by vibrations in the air that create a rise and fall of air pressure. It can be plotted as a simple wave. An analog sound wave follows this original sound wave form exactly, with the intensity of the signal going up and

down continuously with the wave. Digital sound waves are made by repeatedly taking samples of the original wave at a very high rate and converting them into a series of numbers. They are not continuous and have a "staircase" shape.

diaphragm turns sound waves into physical vibrations (and vice versa)

sound recording box

recording stylus, or needle

wax cylinder with grooves for recording

cylinder shaft

cylinder shaft crank

crank handle

Edison Fireside phonograph, 1909

By 1909, Edison phonographs such as this one were a feature of many homes, and there was a wide range of professional wax-cylinder recordings that could be inserted in the machine and played by turning the handle. Unfortunately, wax-cylinders could only be played a few times before the playback quality started to deteriorate.

